# Robert Xinyuan Chi

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## Education

M.A. in Statistics and Data Science, Washington University in St. Louis — Expected May 2027

*Selected Coursework:* Theory of Statistics I–II (PhD sequence: measure-theoretic probability, decision theory, asymptotic theory, and empirical processes), Advanced Linear Models, Statistical Computation, Topics in Machine Learning, Bayesian Statistics, Stochastic Processes, Causal Inference

B.A. in Economics and Mathematics, New York University Abu Dhabi (NYUAD) — May 2025

Study Away at New York University, New York (Spring 2024)

*Selected Coursework:* Honors Real Analysis I & II, Mathematical Statistics, Advanced Econometrics, Applied Optimization, Time Series and Forecasting

## Research Interests

Bayesian methodology, uncertainty quantification, functional data analysis, conformal prediction, statistical machine learning

## Manuscripts

Feldman, J., Bravo, M. A., **Chi, R. X.**, & Kowal, D. R. *Bayesian Decision-Analytic Quantile Regression for Linking Ambient Air Pollution and Longitudinal Academic Performance in North Carolina Children: Adverse Effects of PM2.5 Exposure are Stronger for Lower-Performing Students.* **Submitted to *Data Science in Science*.**

## Research and Professional Experience

**Research Assistant, Washington University in St. Louis**

*Supervisor: Prof. Mengxin (Maxine) Yu* — June 2026 – Present

- Developing a group-theoretic reformulation of sequential conformal prediction methods, expressing permutation-based procedures through group actions and symmetry arguments
- Deriving connections between permutation-based prediction sets and conditional orbital probabilities, with particular focus on conditioning events and finite-sample coverage guarantees

**Research Intern, Institute for Informatics, Data Science, and Biostatistics ($I^2$DB), Washington University School of Medicine**

*Supervisor: Prof. Ruijin Lu* — May 2026 – Present

- Developing and evaluating mimBART, a nonlinear scalar-on-function regression framework using Bayesian Additive Regression Trees to model flexible relationships between functional predictors and scalar responses
- Implementing and comparing FPCA- and B-spline-based functional index estimation, including a shared FPCA representation for fair benchmarking against competing functional regression methods
- Designing simulation studies with 100 replications to assess functional index recovery and out-of-sample predictive performance across multiple sample sizes

**Research Assistant, Washington University in St. Louis**

*Supervisor: Prof. Joseph Feldman* — Jan 2026 – Present

- Studied and implemented Bayesian decision-analytic quantile regression, including posterior summarization, uncertainty quantification, and quantile-specific subset selection

- Designed and conducted Monte Carlo simulations for longitudinal Bayesian quantile regression across 200 simulated datasets and sample sizes up to 5,000, evaluating estimation and calibration under different sample sizes and quantile levels
- Compared the proposed method with competing quantile regression approaches using out-of-sample check loss, quantile calibration, and variable-selection performance metrics

**Research Assistant, Topics in Labor Economics and Microeconometrics**
*Supervisor: Prof. John C. Ham* May 2024 – Dec 2024

- Cleaned, organized, and analyzed approximately 900,000 labor-market observations involving unemployment rates, wages, and education levels from U.S. Department of Labor datasets
- Conducted empirical analysis using regression and difference-in-differences techniques in labor economics and microeconometrics with a focus on labor supply modeling

**Research Assistant, NYUAD Center for Interacting Urban Networks (CITIES)**
*Supervisors: Prof. Kinga Makovi and Prof. Thomas Marlow* May 2023 – Dec 2023

- Developed a database containing over 100 million entries to analyze ESG trends and institutional investment behavior
- Cleaned, merged, and standardized over 20 datasets containing corporate earnings call transcripts and institutional shareholder investment records
- Built a large-scale textual and network dataset with approximately 100,000 ESG and climate-risk observations using named entity recognition, topic modeling, and sentiment analysis
- Analyzed variation in ESG investment behavior across industries and major institutional shareholders based on the constructed dataset

## Technical Skills

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| Languages: | R, Python, SQL, LaTeX |
| Libraries/Tools: | Stan, dbarts, TensorFlow/Keras, scikit-learn, fdapace |

## Honors

- Four-Year Merit Scholarship, NYU Abu Dhabi, 2021–2025
- National 2nd Place in Abu Dhabi National Oil Company (ADNOC)-Bloomberg Energy Investment Competition, 2023